NAME - HARSH RANJAN
ROLL - UG/04/BTCSE/2025/069
SEC - A

PAGE NO.
Date: / /

linked list test

1. Ans → (b) Data and pointer to next node
2. Ans → (c) NULL
3. Ans → (b) Self referential structure
4. Ans → (b) head == NULL
5. Ans → (b) newnode→next = head; head = newnode;
6. Ans → (b) temp = head; head = head → next free(temp);
7. Ans → (c) Traversing free from tail to head.
8. Ans → (b) n
9. Ans → (c) Copying the data of p→next into p, then deleting p→next.

10. Ans → (b) while (p != NULL)

NAME - HARSH RANJAN
ROLL - UG/04/BTCSE/2025/069
SEC-A

11. Ans → (b) p→next = NULL
12. Ans →
13. Ans → (c) Three
14. Ans → (b) Exactly one node
15. Ans → (b) (struct Node*) malloc (size of (struct Node))
16. Ans → (b) Avoid special-case handling for insertion and deletion

17. Ans → (c) The first node
18. Ans → (b) It does not support random (linked list) access
19. Ans → (b) Loss of access to all
20. Ans → (b) A slow pointer and one fast pointer
21. Ans → (c) The last node
22. Ans → (b) Nodes may be scattered anywhere in memory
23. Ans →

```
struct Node {
    int data;
    struct Node *next;
};
```

NAME = HARSH RANJAN
ROLL = UG/04/BTCSE/2025/069

Teacher's Signature.....

NAME - HARSH RANJAN
ROLL - UG/04/BTCSE/2025/069
SEC - A

PAGE NO.
Date: / /

24. Ans → 30, 40, 70

28 (i) Ans →

```
Struct node * Insert At Beginning
    (struct Node * head, int data) {

Struct Node * ptr = (struct node *) malloc
    (sizeof (struct Node));

    ptr -> next = head;
    ptr -> data = data;

    return ptr;

}

head = insert At beginning (head, 7);
linked list Traversal (Head);

    return 0;
}
```

(iii) Ans →

```
Struct Node * & Insert at End (Struct node * head,
                                int data) {

struct Node * ptr = (struct Node *) malloc (sizeof
                    (struct node));

Struct Node * p = head;
    p = p -> next;
    while (p -> next != null) {
}       p = p -> next

    p -> next = ptr;
    ptr -> next = NULL;
    ptr -> data = data;

    return head;
}
```

NAME = HARSH RANJAN
ROLL = UG/04/BTCSE/2025/069
SEC = A

Teacher's Signature.....

PAGE NO.
Date : / /

```
head = InsertAtend (head, 9);
linkedlistTraversal (head);

return 0;
}
```

NAME = HARSH RANJAN
ROLL = UG/04/BTCSE/2025/069
SEC = A

Teacher's Signature